VYORA

Small Business Billing, Inventory & Management Platform

Document: Strategic V1 Architecture & Roadmap Brief
Release Track: Version 1.0 (Production Core)
Target: Indian MSMEs, Retailers & Wholesalers
Updated: October 2026

1. Executive Summary & Core Product Architecture

Vyora is an all-in-one cloud billing, receivables, supplier payables, and traceable inventory management platform purpose-built for Indian micro, small, and medium enterprises (MSMEs). Our core mission is delivering **data security, rapid user adoption, Theme B (Teal Ledger) visual clarity, and flawless concurrency stability** (1,000 to 50,000+ tenants) without over-engineering.

vyora-backend (REST API Core)

NESTJS 11 + MONGOOSE 8

Stateless modular API. Enforces tenant boundaries via `TenantGuard`, role controls via `RolesGuard`, and quota limits via `PlanGuard`. Minor currency integer precision across ledgers.

vyora-frontend (Web Admin App)

REACT 19 + SHADCN/UI + VITE 6

Responsive SPA built on Theme B (Teal Ledger), Radix UI popovers, shadcn/ui Table suite, and Toast feedback. Debounced search, edge-to-edge modal footers, and icon-only table actions.

vyora-website (Marketing & Tools)

PORT 3001 • 10 MSME TOOLS + SEO

High-converting marketing hub featuring live pricing plans, SEO blog articles, and 10 interactive utility tools (QR generator, GST calculator, currency converter, HSN finder, loan EMI).

vyora-mobile (Mobile POS & Billing)

FLUTTER / REACT NATIVE MONOREPO

Upcoming cross-platform counter app consuming the **exact same NestJS backend and APIs** via shared types, Bearer JWT authentication, and local offline cache.

2. Roles & Rights: Analysis of the Demotion Issue & Clean Resolution

In the recent team management audit (referenced in user incident screenshot `media_1791700339539_0dcc8fac.png`), a business owner attempting to demote a secondary team member from **Owner** to **Staff** encountered the blocking error:

> **Observed Error:** `"Cannot demote the last remaining owner. Assign another owner first."`
> *Context:* The business had 2 total members — both marked as Owners (Rahul Sharma as "Primary Owner" and Sandip Yadav as "Owner"). When Rahul Sharma toggled Sandip Yadav to Staff, the request was rejected.

Root Cause Analysis

- **Collection Query Disconnect:** The backend rule in `business.service.ts` checked `membershipModel.countDocuments({ businessId, role: 'OWNER', isActive: true })`.
- **Legacy & Undefined Field Exclusion:** In MongoDB, querying `{ isActive: true }` strictly filters for documents containing explicit boolean `true`. Seeded records or memberships created before the schema update had `isActive: undefined`, causing MongoDB to count only 1 active owner instead of 2.
- **Primary Owner vs Promoted Owner Disregard:** The database stores the root account creator inside `business.ownerUserId`. The primary owner is permanently immune to role changes. Therefore, even if all secondary owners are demoted to Staff, the Primary Owner *always* remains, guaranteeing that a business is never left without an owner.

The Robust V1 Invariant Rule & Architecture Fix

User Type	Immutability Status	Promotion / Demotion Permissions
Primary Owner `business.ownerUserId`	PERMANENTLY IMMUNE	• Can promote any Staff member to Owner anytime. • **Can demote any promoted Owner back to Staff anytime** without triggering the "last owner" block. • Cannot demote or delete their own primary account.
Promoted Owner `role: 'OWNER'` in Membership	FULL WORKSPACE ADMIN	• Full access to invoices, catalog, purchases, payments, and reports. • Cannot demote or modify the Primary Owner. • Cannot demote other promoted Owners (only Primary Owner holds demote rights).
Operational Staff `role: 'STAFF'`	CONFIGURABLE ACCESS	• Subject to module visibility matrix and action rights (View, Create, Edit). • Cannot access team management, subscription upgrades, or company deletion.

3. Re-Organized Flow: Consolidated Account & Workspace Hub

Currently, user information, team permissions, and billing actions are dispersed across multiple screens. In Version 1, we organize these into a unified, elegant **"Account & Workspace"** hub with intuitive nested tabs.

1. Roles & Permissions NESTED TAB A

- Team member invitations (Email/Phone).
- Role Switchboard (Promote / Demote).
- Custom Role Builder with module visibility.
- Active / Suspended access toggle.

2. Billing & Subscription NESTED TAB B

- Current Plan badge (Starter, Growth, Pro).
- Usage meters (Invoices, Staff, Storage).
- Secure checkout with dynamic UPI QR.
- GST invoice history for tax credit.

3. Business & Profile NESTED TAB C

- Legal Name, Trade Name, GSTIN, PAN.
- Billing address & State code (POS rules).
- Bank details & UPI ID for invoice QR.
- Company Logo & Digital Signature.

4. Custom Roles & Granular Module Visibility Matrix

Small businesses require role flexibility without technical complexity. Vyora provides clean **Module Visibility (Hide/Show)** coupled with **Action Toggles (View, Create, Edit, Delete)** per module.

Module	Owner / Admin	Accountant (Custom)	Billing Operator (Custom)	Stock Incharge (Custom)
Invoices & Billing	FULL CRUD + CANCEL	VIEW + EXPORT	CREATE + PRINT ONLY	MODULE HIDDEN
Products & Catalog	FULL CRUD	VIEW PRICES	VIEW ONLY	FULL STOCK EDIT
Customers & Dues	FULL CRUD	FULL LEDGER VIEW	VIEW & SELECT ONLY	MODULE HIDDEN
Suppliers & Purchases	FULL CRUD	FULL CRUD	MODULE HIDDEN	RECORD INWARD STOCK
Payments & Receipts	FULL CRUD	FULL CRUD	RECORD RECEIPT ONLY	MODULE HIDDEN
Tax Reports & Analytics	FULL ACCESS	FULL ACCESS (CA/TAX)	MODULE HIDDEN	MODULE HIDDEN
Account & Roles	OWNER ONLY	MODULE HIDDEN	MODULE HIDDEN	MODULE HIDDEN

5. Competitor Strategy: How Vyapar Manages Roles & What Vyora Does Better

How Competitor "Vyapar" Operates DESKTOP-FIRST LEGACY

- **Role Presets:** Offers Admin, Salesman, Biller, CA/Accountant, and Delivery Boy.
- **Binary Restriction Checkboxes:** Uses simple toggles: *"Can see purchase prices? NO"*, *"Can edit old invoices? NO"*, *"Can give discounts? NO"*.
- **Local-First Sync:** Runs on a local desktop database that syncs to cloud drive, leading to sync conflicts when multiple devices write simultaneously.

Vyora's Enhanced V1 Strategy CLOUD-NATIVE MODERN

- **Zero Sync Conflicts:** Real-time cloud persistence with optimistic locking on MongoDB Atlas.
- **Module Hiding:** If an operator doesn't need Reports or Purchases, the navigation link completely disappears — zero distraction or confusion for staff.
- **Instant Role Switchboard:** 1-click active/suspend and role switch toggles without requiring manual app re-installation or complex permissions wizards.

6. Billing Management & Secure UPI QR Code Flow

Billing occurs across two distinct domains: **Subscription Billing** (tenants paying Vyora) and **Customer Invoicing Billing** (buyers paying tenants). Both must adhere to secure, standardized payment flows.

A. Subscription Checkout & Missing QR Fix

> **Resolution for Subscription Modal UPI:** When users select UPI/QR in `PlanPurchaseModal.tsx`, instead of showing a blank state, the system renders a dynamic Bharat UPI QR code encoded with merchant VPA, exact plan amount in minor units, and unique idempotency reference.

- **Standard Indian VPA Format:** `upi://pay?pa=vyora.billing@icici&pn=Vyora%20Cloud&am=2999.00&tr=TXN_98231&cu=INR`
- **Production Payment Gateway (Roadmap):** Integration with Razorpay / Cashfree Standard Hosted Checkout for automatic webhook reconciliation, recurring renewals, and automated GST input tax credit (B2B Tax Invoice generation for the subscriber).

B. Counter Invoicing Dynamic UPI QR Code

- **Per-Invoice Dynamic QR:** When a counter cashier creates a sales bill, the app displays a dynamic QR code on screen or prints it on thermal/A4 invoices.
- **Automatic Settlement Tagging:** Customer scans with Google Pay, PhonePe, or Paytm. Cashier clicks "*Payment Received via UPI*" to generate an idempotent payment voucher and mark the invoice paid.

7. URL Slugs: Evaluation & Strategic Recommendation

The user enquired: "*also i think to manage the slug in our app that was a good for us or not so think about that*". Here is the architectural analysis:

Approach	Pros	Cons & Recommendation
Option A: Slugs in App URLs `app.vyora.in/:slug/invoices`	• Clear tenant identity in browser tab. • Accountants can open multiple client tabs side-by-side.	**COMPLEX & FRAGILE FOR V1** • Routing breaks if tenant renames trade name. • Route collision risk (e.g. business named "settings" or "reports"). • Requires route guard checking slug ownership on every click.
Option B: Clean URLs + Tenant Switcher `app.vyora.in/invoices` (JWT Context)	• Clean, rock-solid, bookmarkable URLs. • Business context managed via JWT + dropdown. • **Used by Vyapar, Khatabook, Zoho Books, and Stripe.**	**STRONGLY RECOMMENDED FOR V1** • Zero frontend routing bugs. • Frictionless multi-business switching from navbar.
Option C: Public Slugs for Customers `vyora.in/pay/:invoice-slug`	• Customers can view and pay invoices online. • Future online catalog storefront: `vyora.in/store/:slug`.	**PERFECT USE CASE FOR SLUGS** • Keeps public links human-readable while keeping internal app clean and robust.

8. Cross-Platform Mobile App: Architecture Consuming the Same Backend

The user requested: "*i want to create a seprate mobile app and want to use the same backend and api so how we can manage this this is possible or not*".

> **Architectural Verdict: 100% Possible and Recommended.** The Vyora NestJS backend is completely decoupled, stateless, and REST-compliant. Both Web and Mobile consume the exact same backend endpoints without code duplication.

Shared Monorepo Foundation

`@VYORA/SHARED-TYPES`

Data models (`Invoice` , `Product` , `Party` , `PaymentRecord`) and validation schemas live in shared monorepo packages. Mobile app directly imports the same contracts.

Authentication & Device Security

`JWT BEARER + REFRESH TOKEN`

Mobile uses the same `/auth/login` endpoint. Access tokens stored in EncryptedSharedPreferences (Android) / Keychain (iOS) with biometric unlock (Fingerprint / Face ID).

Offline Counter Billing (Inspired by Vyapar)

`HIGH-VALUE MOBILE CAPABILITY`

Indian retail counters frequently experience spotty internet. By integrating a local SQLite / WatermelonDB cache on the mobile device, cashiers can generate bills offline. Bills queue locally and automatically sync to MongoDB via idempotent endpoints (`idempotencyKey`) as soon as Wi-Fi/4G reconnects.

9. Subscription Tiers & Feature Gating Matrix

All tenants register on the **Starter (Free Forever)** tier upon signup. As business volume scales, upgrade gates gently nudge owners toward Growth and Business Pro tiers.

Feature Module	Starter (₹0 / Free)	Growth (₹2,999 / yr)	Business Pro (₹4,999 / yr)
Sales Invoicing	50 INVOICES / MONTH	UNLIMITED GST BILLS	UNLIMITED + BARCODE
Product Catalog & Stock	UP TO 100 PRODUCTS	UNLIMITED + LOW STOCK ALERTS	UNLIMITED + MULTI-UNIT HSN
Party Directory & Balances	BASIC BALANCES	FULL LEDGERS & CREDIT LIMITS	FULL LEDGERS & REMINDERS
Purchases & Payables	HIDDEN (UPGRADE GATE)	INBOUND SUPPLIER BILLS	FULL INBOUND BILLS
Payments & Vouchers	HIDDEN (UPGRADE GATE)	RECEIPTS & PAYMENT VOUCHERS	FULL CASH/BANK VOUCHERS
Tax Reports & Analytics	HIDDEN (UPGRADE GATE)	GSTR-1, 3B, VALUATION	FULL CA TAX EXPORT
Team Staff Delegation	1 SOLO OWNER ONLY	UP TO 3 STAFF ACCOUNTS	UNLIMITED CUSTOM ROLES
Audited Stock Adjustments	HIDDEN	HIDDEN	FULL AUDITED ADJUSTMENTS

10. Concurrency Stability, Security & Data Protection (1k - 50k Users)

Indian billing platforms must never crash during month-end or festive shopping surges. Vyora integrates enterprise stability controls:

- **Sliding-Window Rate Limiting (`rate-limit.middleware.ts`):** Restricts auth endpoints to 25 req/min (anti-brute-force) and general API endpoints to 200 req/min. Sweeps memory every 10 minutes to support 50k unique IPs without leaking heap.
- **Client Search Debouncing (`SearchInput.tsx`):** 300ms debounce eliminates query hammering while users search products, invoices, or parties.
- **High-Capacity Data Table Architecture (`<Pagination>` + `Table.tsx`):** Virtualized page chunks (10, 25, 50, 100) ensure zero browser freezing even with 50,000+ customer records.
- **Zero Floating-Point Ledger Precision:** All currency stored in minor integer units (paisa) prevents rounding drift across multi-line tax calculations.

11. Actionable Strategic Suggestions for Vyora V1 Launch

1. WhatsApp Invoice Dispatch — HIGH DEMAND IN INDIA

Indian retailers love sending bill PDFs directly over WhatsApp. Add a 1-click `wa.me/?text=...` button on invoice finalization with pre-filled customer message and link.

2. Barcode Scanner Support (USB/Camera) — ESSENTIAL FOR RETAIL

Enable automatic product lookup when a USB barcode scanner enters a SKU code into the invoice item row, speeding counter checkout by 5x.

3. Thermal Printer (2-inch / 3-inch) Support — POS RECEIPT FORMAT

In addition to regular A4 GST tax invoices, provide a compact 80mm thermal receipt print stylesheet for grocery stores and small retail shops.

4. Daily Auto-Backup to Google Drive — DATA PEACE OF MIND

Allow owners to download encrypted JSON/Excel backups of their full ledgers with a single click, providing absolute data sovereignty.